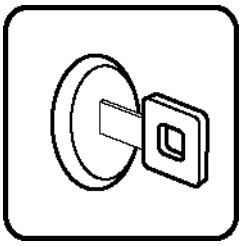
Start Up and Shut Down Procedures



Correct use of start up and shut down procedures are important to ensure continued high productivity and safe operation.

This chapter includes the following sections for starting up and shutting down the machine and temperature control:

- Start Up from a Cold Condition or a Power Failure
- Temporary Cool Down
- Shut Down
- Setting Up Temperature Control Screens
- Running Purge Material

Note: *Some functions are optional and may not be configured on all machines.*

Note: *Before Starting the System - The setpoints are preset to meet customer specifications before delivery. The setpoints are held in memory.*

Start Up from a Cold Condition or a Power Failure

Starting the machine requires preheating and melting the resin in the extrusion system before the molding process can begin.

Preheating and “soak” will take approximately 2-1/2 hours.

The Start Up procedures include:

- Inspect idle machine and auxiliary equipment.
- Apply utilities and heat.
- Turn on auxiliary equipment.
- Prepare machine for start up.
- Prepare blowpins, if necessary (optional function).
- Run material.

1. Inspect the Idle Machine and Auxiliary Equipment

Before starting the machine, inspect for any signs of equipment damage or failure. Correct any problems before starting the machine.

- Inspect for water or coolant leaks on reservoirs, hoses, and fittings.
- Remove any containers that remain in the machine.
- Ensure that there is a supply of resin and the hopper is full.
- Look for any loose hardware.
- Look for bolts, nuts, or other fasteners that may have come loose.

⚠WARNING

Do not start the machine until all conditions have been corrected.

2. Apply Utilities and Heat

a. Turn water and air on.

b. Check for any leakage.

c. Turn main electrical disconnect On. This is located on the electrical cabinet.

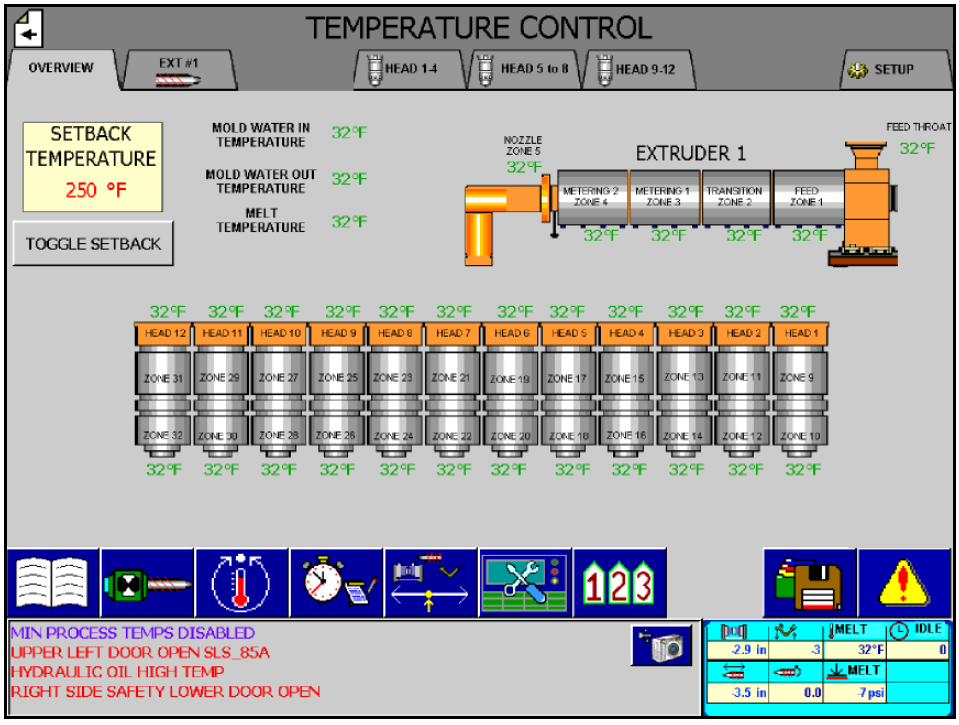
d. Turn Control Power On from the Operator Console.

e. Press  on the menu bar to access the Temperature Control screen. The screen will display the setpoint and actual temperatures for all zones. Verify correct setpoints for resin type. This procedure will need the operator to be logged in to change any setpoints. The Temperature Control screen allows the user to select what area of the machine to view/change heat control and alarming information.

f. Press the Heat Control button on the Operator Console. This will enable the machine to reach preset temperature. The machine will reach full preset temperature within 60-90 minutes. The machine should then "SOAK" for an additional 30 minutes to ensure the resin has completely melted.

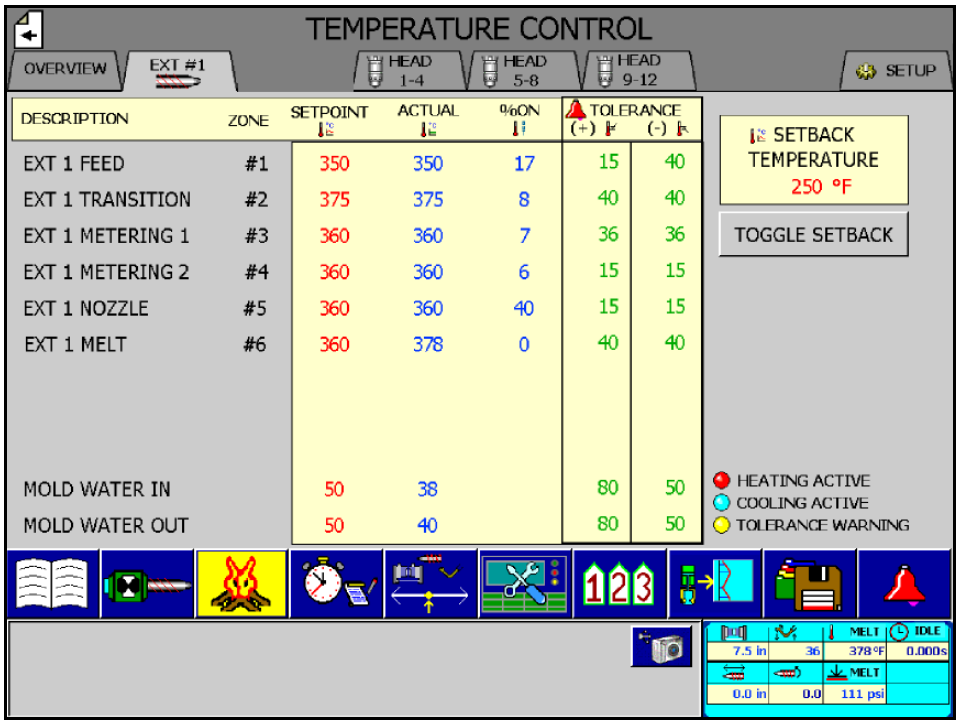
Note: *The Operator Console for the Non-Parison Programming is the same, but without the Programmer Hydraulic Pump button.*

Note: *If the machine is not properly "SOAKED" at temperature, a "COLD START of the Extruder can cause screw breakage.*

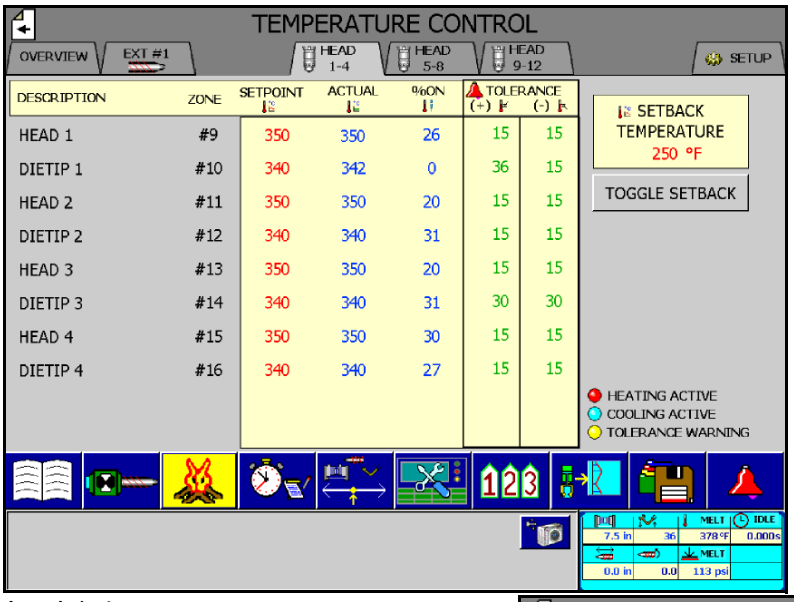


Temperature Control Screen

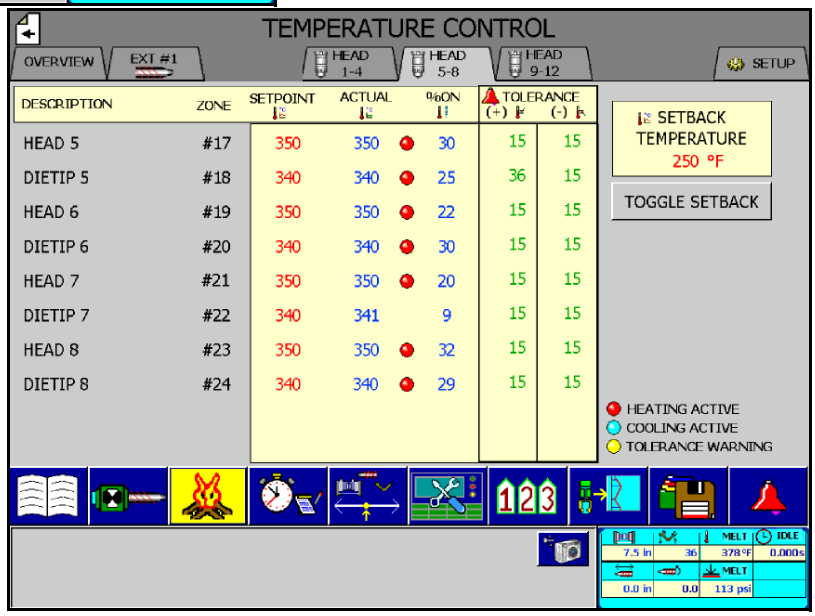
The Extruder #1 Temp. Zones 1-8 screen allows the user to view current temperature and change temperature zone information.



Extruder #1 Temp. Zones 1-8



Head 1-4



Head 5-8

Temperature Control - Heads 1-8

Once Run Temperature is reached, it may be desired to activate the SETBACK TEMPERATURE if the machine will not be in use for an extended period.

Note: *SETBACK TEMPERATURES ENABLED is indicated by a background that turns red.*

The Temperature Control - Trend screen allows the user to select a heat zone and watch how close the temperature is tracking with the setpoint.

3. Turn On Auxiliary Equipment

Approximately 15 minutes before production (Auto Cycle), start auxiliary equipment.

a. Press the External Circuit button on the Operator Console of the Blow Molder.

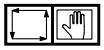
b. Press the Control Power button on each Operator Console of each piece of auxiliary equipment.

⚠ CAUTION

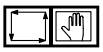
After turning on each device, visually inspect for any discrepancies.

4. Prepare Machine for Start Up

a. Ensure that all temperature controllers are operating normally. All displays should show controllers cycling properly. If operation is not correct, contact a service technician.

b. On the non-operator side of the machine, switch  to  . Only one side can be active.

c. Press the Hydraulic On button. Visually check the machine for leaks.

d. On operator side of machine, switch to  to manual.

e. Close all doors and reset safety door circuit (Yellow lighted button).

f. Run Swingarms up and down several times to verify operation and circulate the oil. Leave the arms in down position.

g. Clean plastic from Blowpins, Dies and Mold and run blowpins (if equipped) up.

⚠ CAUTION

To prevent damage to blowpins, do not close molds fully.

h. Run Clamp open and closed several times to verify operation and circulate oil. Leave Clamp in open position.

5. Prepare Blowpins (Optional)

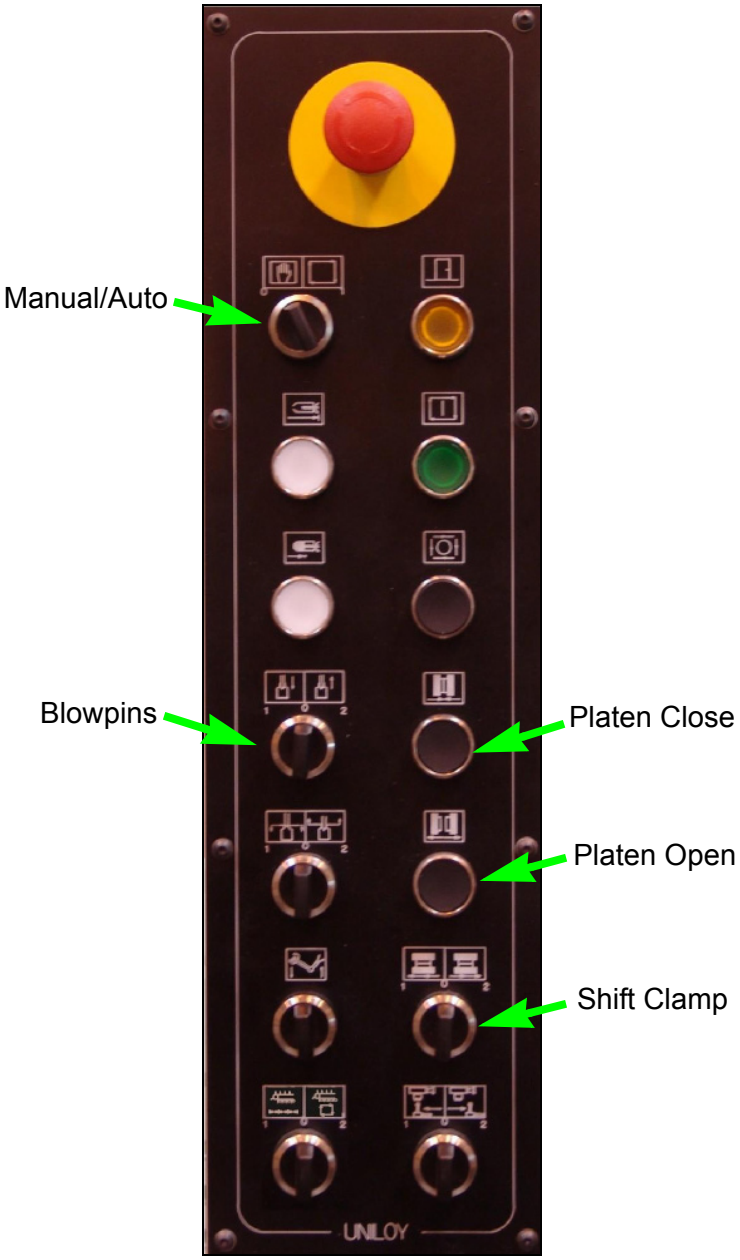
The Blowpins may need to be cooled if the parison diameter is close to the blowpin diameter. The plastic may stick to the blowpin and not drop consistently enough to start the machine.

Another situation to cool the blowpins in with ram down prefinish. The items will be hotter than in the run condition and may not trim too well at the start.

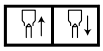
a. Jog molds closed until they are 6mm (.25 inch) from full close (jog on Operator Station).

⚠ CAUTION

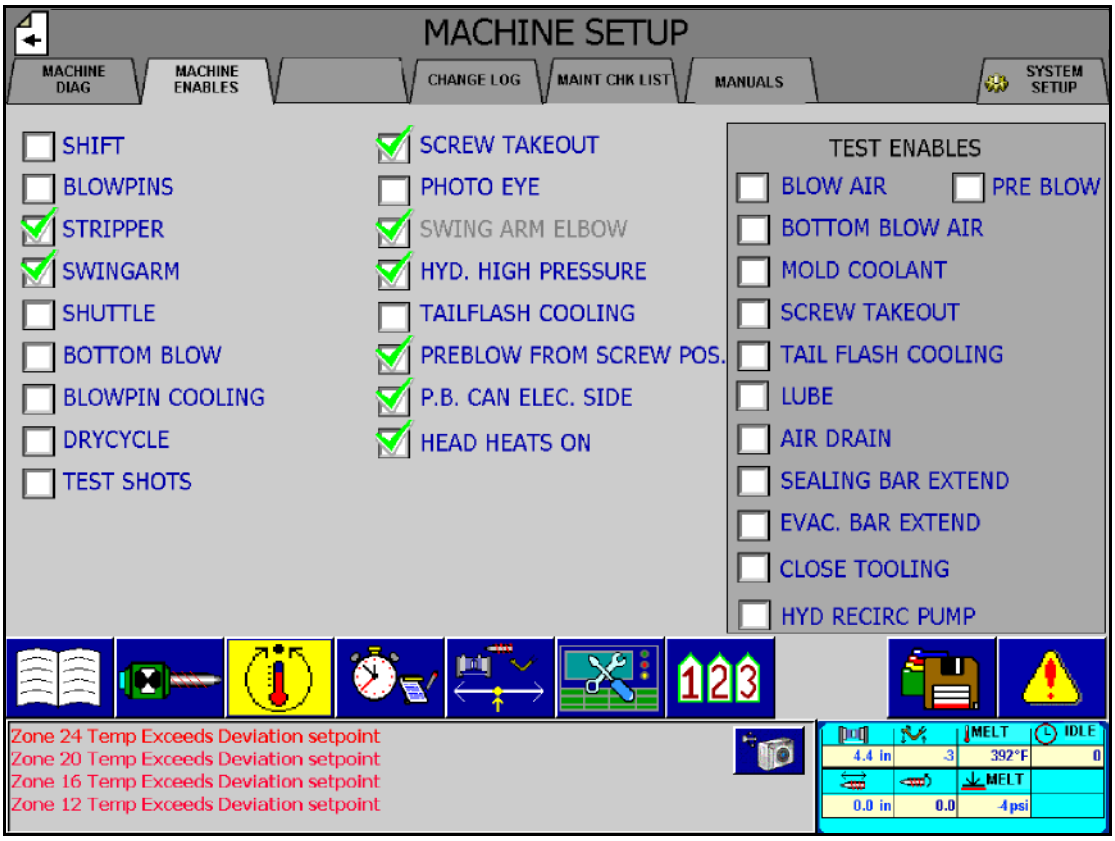
To prevent damage to blowpins, do not close molds fully.



Operator Station (R2000 shown)

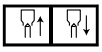
b. Move the Blow Pin down by switching  to  on the Operator Station.

c. Touch  on the menu bar then press the MACHINE ENABLES on the Machine Setup screen. Press BLOW AIR to turn On for 10 to 15 seconds, then press BLOW AIR again to turn Off.



Machine Setup - Machine Enables Screen

Note: *Do not leave the blow air on longer than 15 seconds. If left on too long the air will start to cool the die heads.*

d. Move the Blow Pin up by switching  to  .

e. Open the molds by pressing  on the Operator Station.

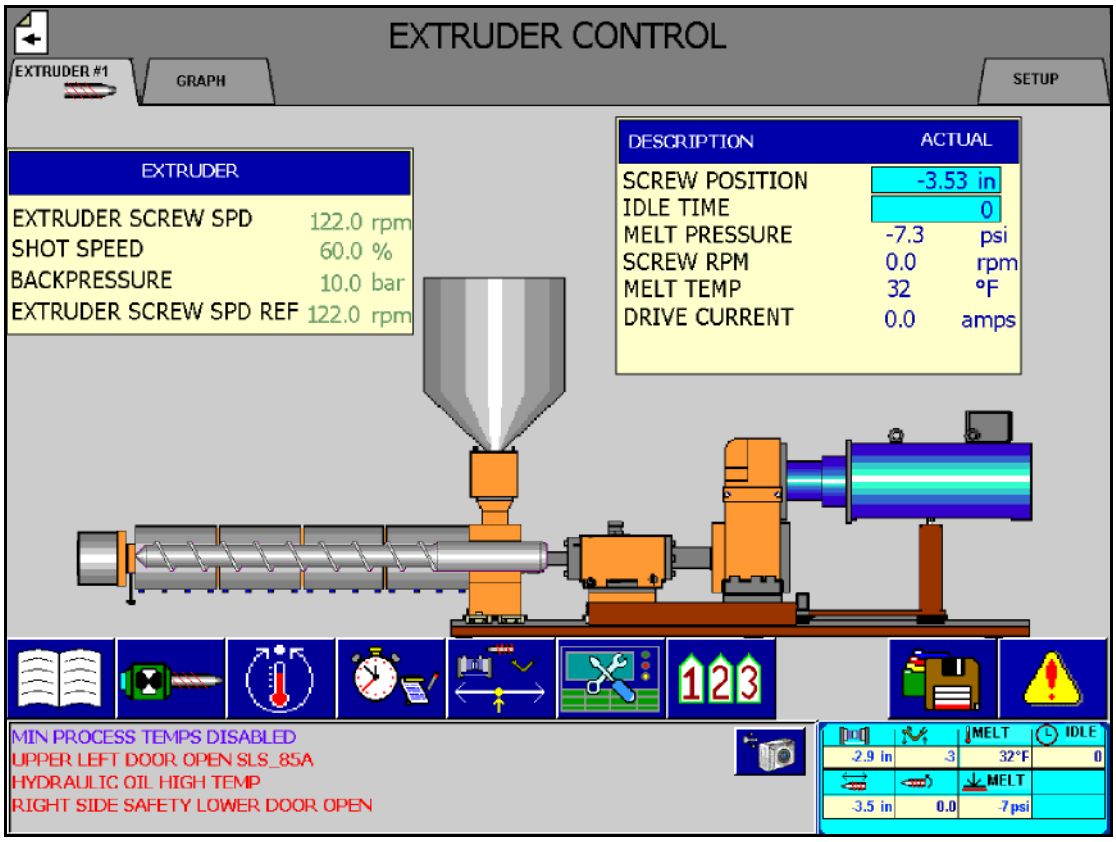
6. Run Material

Start the extruder:

Observe Melt Temperature on the Extruder Control screen. Make sure it and all other temperatures are at setpoints per earlier directions.

a. Ensure  is turned to Divert.

b. Press  on the menu bar. This screen displays the extruder control parameters.



Extruder Control - Extruder Speed Screen

Note: *Ensure that the final screw speed is entered in the "Extruder Screw SPD REF box or the speed setting is recorded for future use.*

The Extruder Screw SPD REF does not control the extruder speed it is a reference point only. The control stores the EXTRUDER SCREW SPD in the "REF" box anytime the extruder is stopped.

c. Decrease extruder screw speed to 30-40 rpm on the Extruder Control screen.

d. Press the Extruder ON button on the Operator's Console.

e. Plastic should begin to extrude within 30 seconds. If no plastic is extruded, a high output current (amps) or high melt pressure (psi) is observed on the Extruder Control screen, shut down the extruder and contact a technician.

Note: *Air releasing from melt often creates a popping noise. The noise indicates normal operation at start up from a cold condition.*

f. Once plastic is extruded, slowly increase extruder speed to the extruder screw SPD REF.

g. Check to ensure even flow.

h. Hold down the Charge button until shot size light comes on. Then release it and the machine will purge the shot.

i. Repeat step 'g' several times watching for even tail lengths and straight parisons.

Note:  *should still be in Divert.*

j. Touching  will put the machine in Auto Cycle.

k. Switch  to Run after molds close and before the swing arms go up.

Temporary Cool Down

Temperature setback allows the machine to cool for maintenance or other operations. Use setback temperature only when the machine will be left for more than 45 to 60 minutes or when a maintenance procedure will take longer than 45 to 60 minutes to complete. Temperature setback typically reduces machine temperature to approximately 250°F (121°C).

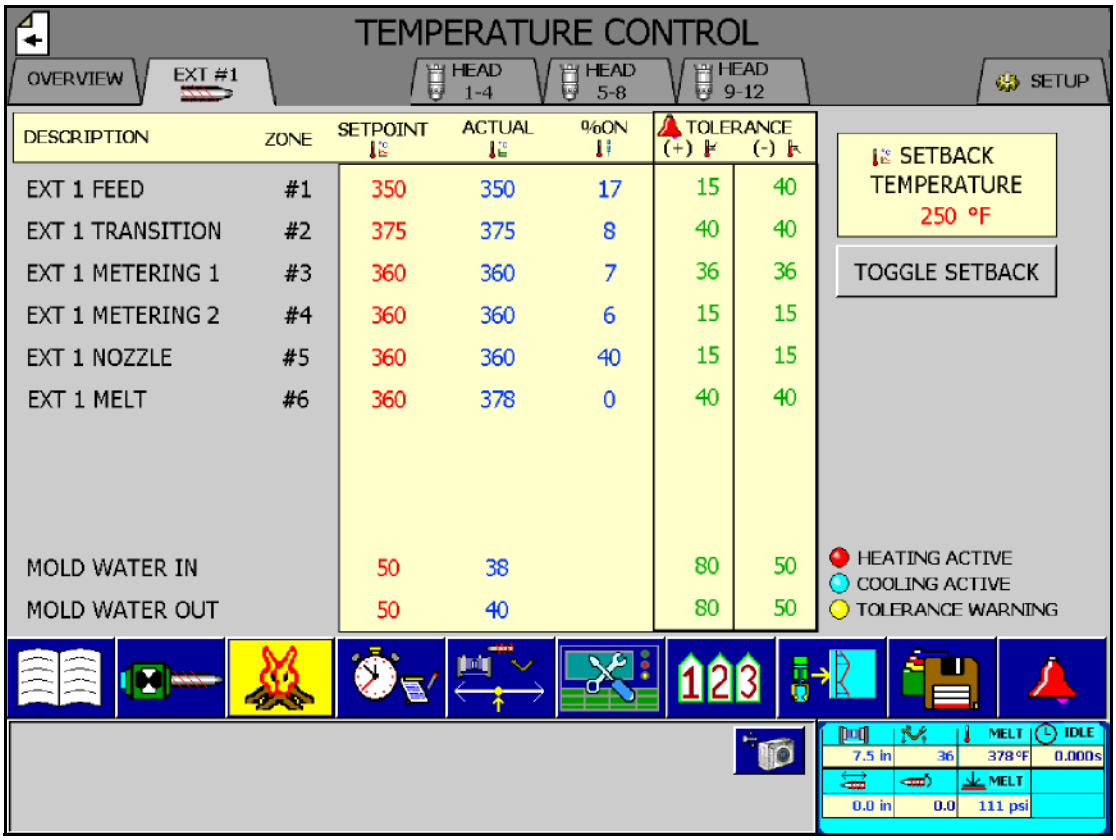
a. Touch  on the menu bar.

b. Select EXTRUDER #1 on the Temperature Control Screen.

c. Touch TOGGLE SETBACK to enable SETBACK TEMPERATURES.

Note: *SETBACK TEMPERATURES ENABLED is indicated by a background that turns red.*

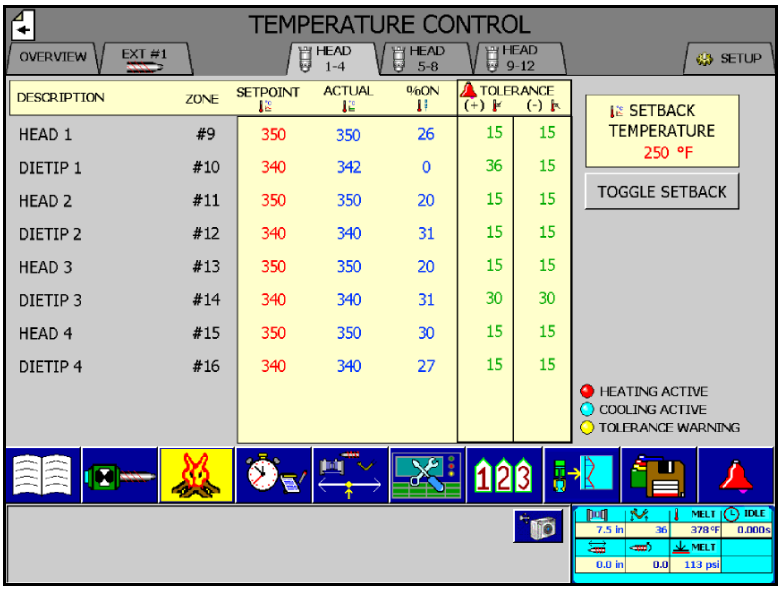
d. Touch TOGGLE SETBACK again to set temperatures back to normal.



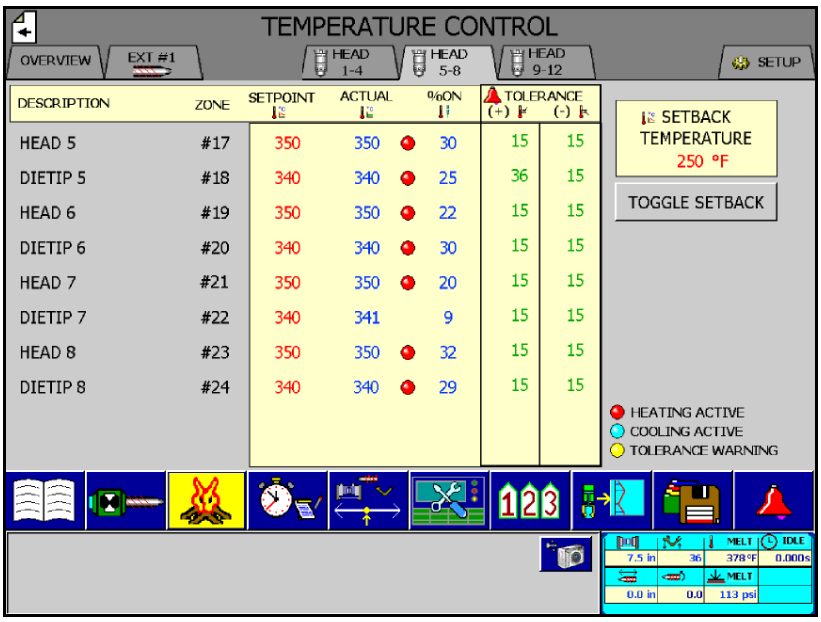
Temperature Control Screen

	Function
SETPOINT	Setpoint is the desired temperature.
ACTUAL	Actual is the feedback.
% On	% On is an indication of how hard the control is working to maintain the setpoint.
(+) TOLERANCE	Temperature warning limit.
(-)TOLERANCE	Temperature warning limit.

The Extruder #1 Temp. Zones 1-8 screen allows the user to view current temperature and change temperature zone information. Select the HEAD tabs to view and change those settings.



Head 1-4



Head 5-8

Temperature Control - Heads 1-8

Shut Down

Careful shut down ensures that the machine can be restarted safely and quickly. These shut down procedures also ensure the best productivity from the machine.

a. Touch  on the Operator Station. This allows the machine to finish the current cycle but not continue into further cycles. It will shut down the extruder motor and the clamp.

b. Toggle SETBACK TEMPERATURE to ENABLED on the Temperatures screen.

c. Wait about 15 minutes. Cooling plastic will try to move extruder slowly back. Leaving hydraulics On will hold the screw fully forward.

d. To turn Off, press Control Power on the Operator Console.

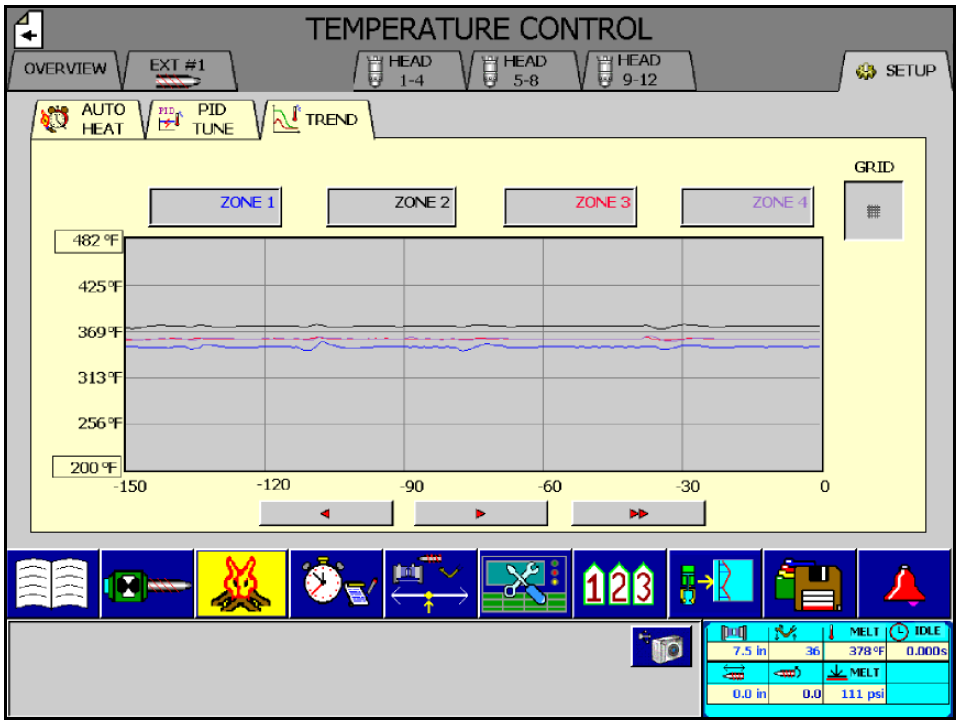
e. After temperatures have cooled below 300°F (150°C) turn main power disconnect Off.

f. Turn water and air off.

Setting Up Temperature Control Screens

Setting Up Trend Screen

The Trend Screen allows the user to select a heat zone and watch how close the temperature is tracking with the setpoint.



Temperature Control - Trend Screen

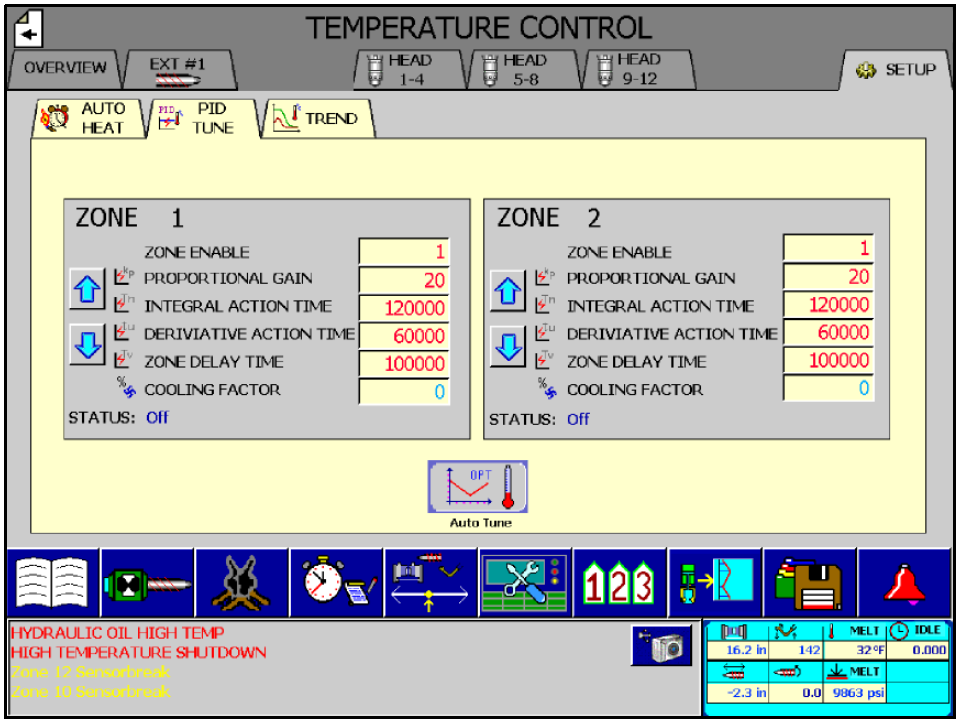
Each of the four boxes can display any of the heat zones on the machine. Touch the box and select the zone.

The grid box turns the grid line On or Off.

To change the scaling of the temperatures, touch the High and/or Low limit boxes to enter a new number. This “screen shot” can be saved to a USB device using the Camera button. The actual data cannot be saved.

Manually Tuning the Machine Temperature Zone PID

An engineer typically adjusts the Machine Temperature Zone PID (Proportional Integral Derivative) to get the best temperature curve appropriate for the process. Temperature zones are defined on the Temperature Control screens.



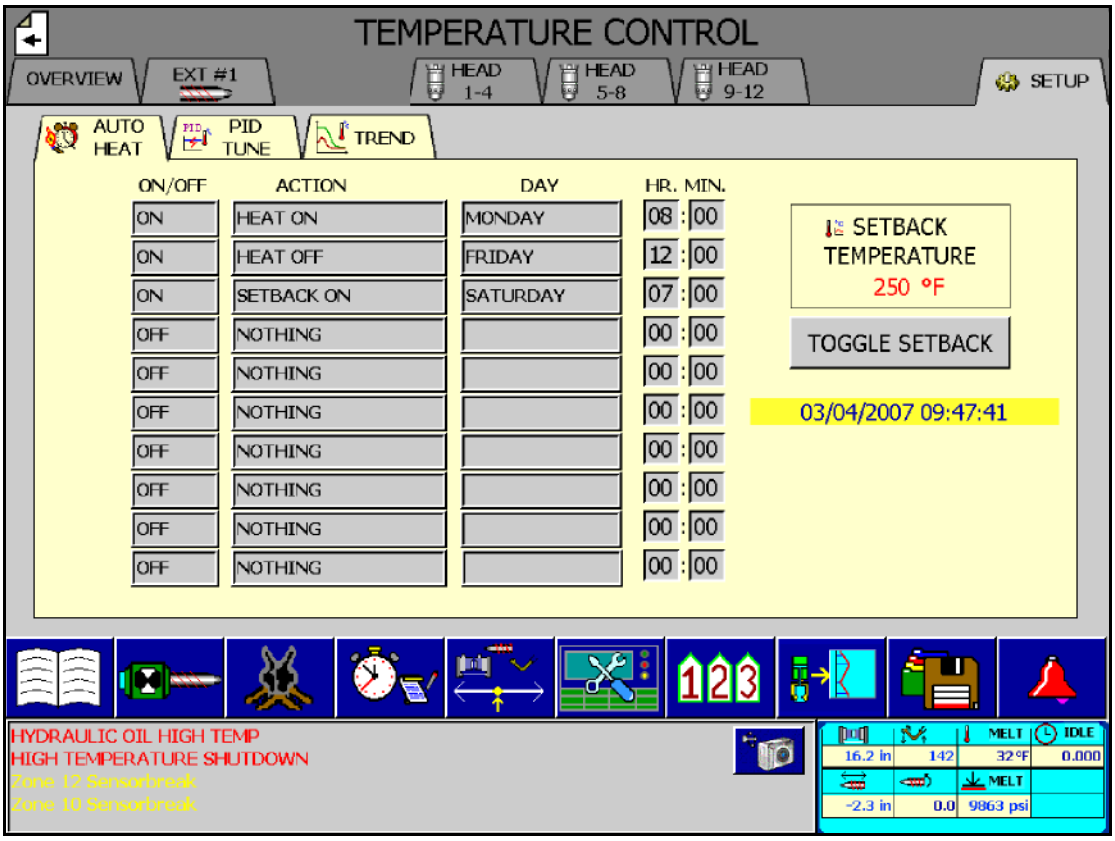
PID Tuning Screen

Auto Heat

The Temperature Control - Auto Heat screen allows the operator to control the heats and setback enables automatically. To enable an ACTION, press On/Off to On. Then select an ACTION. The ACTION choices are:

Button	Function
HEAT ON	Enables heat control.
HEAT OFF	Turn heat control to OFF.
SETBACK ON	Enables setback temperatures
SETBACK OFF	Turns setback temperatures to OFF.
DO NOTHING	No action.
DAY	Sets the Day of the Action.
HR. - MIN.	Sets the time. Time is on 24 hours format.

To automatically turn Heat On in SETBACK, initiate Heat On, one minute later initiate SETBACK On. To return to TEMPERATURE SETPOINTS, after soaking in SETBACK, initiate SETBACK Off.



Temperature Control - Auto Heat

Running Purge Material

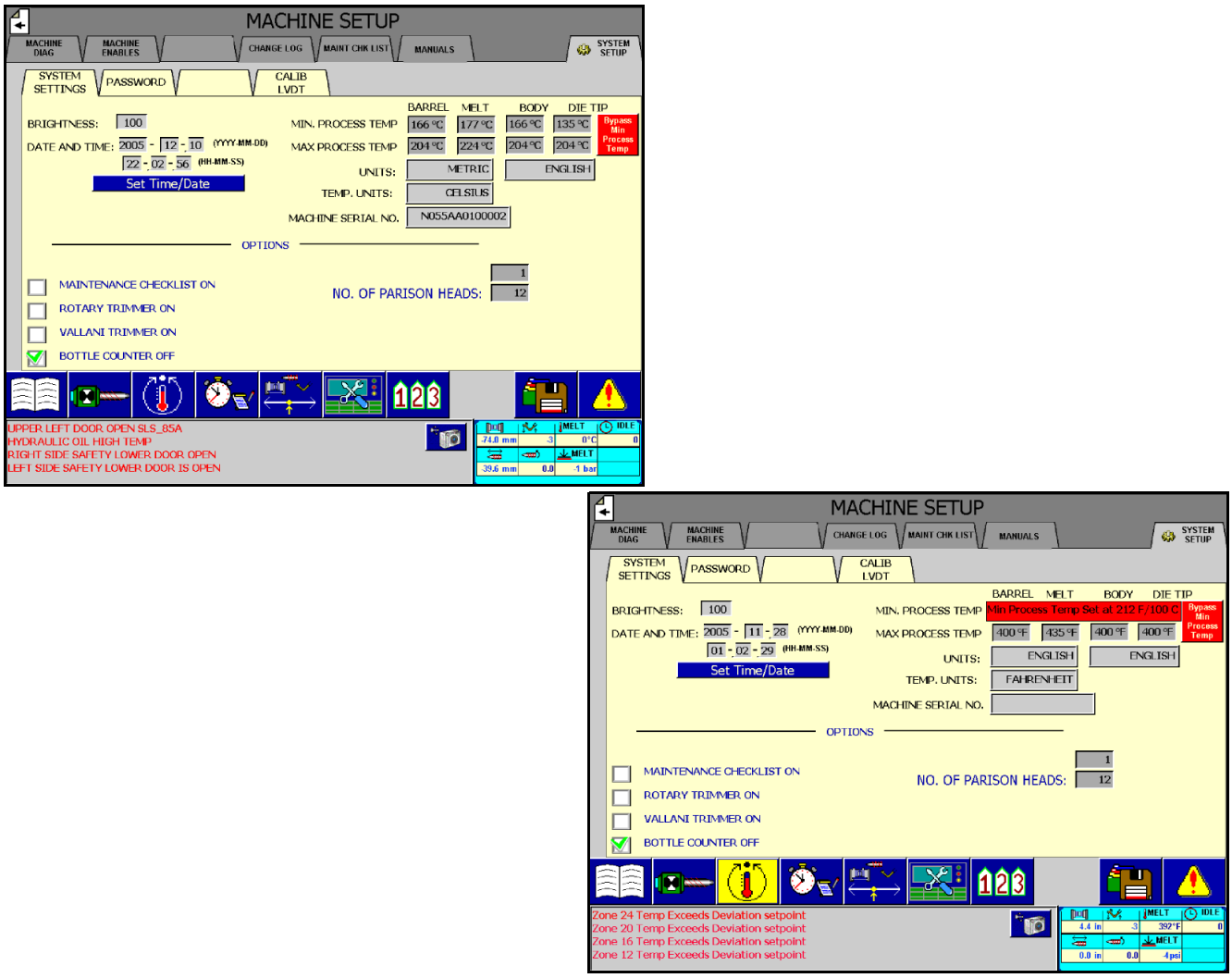
To effectively run some purge materials, the temps must run below the minimum process temperature limits set in the control. Not all purge material needs to run below minimum.

To by-pass these settings, go to Machine Setup screen, then the System Setup tab. Operator must have Admin rights to do this process.

Press the red button 'By Pass Min Process Temp'. This will then re-limit the minimums to 32°F (0°C). With the Min Process Temp bypassed, the Auto Cycle is not functional.

To return to the Auto Cycle Min Process Temps, press the By Pass button again to turn it Off.

The field MACHINE TYPE, is set at the factory and is used to configure the screens for a R2000 or UR65. The values in this box will be either a 2000 for a R2000 machine, or 65 for a UR65.



Machine Setup - Bypassing Minimum Process Temperatures

Control System Calibration and Diagnostics

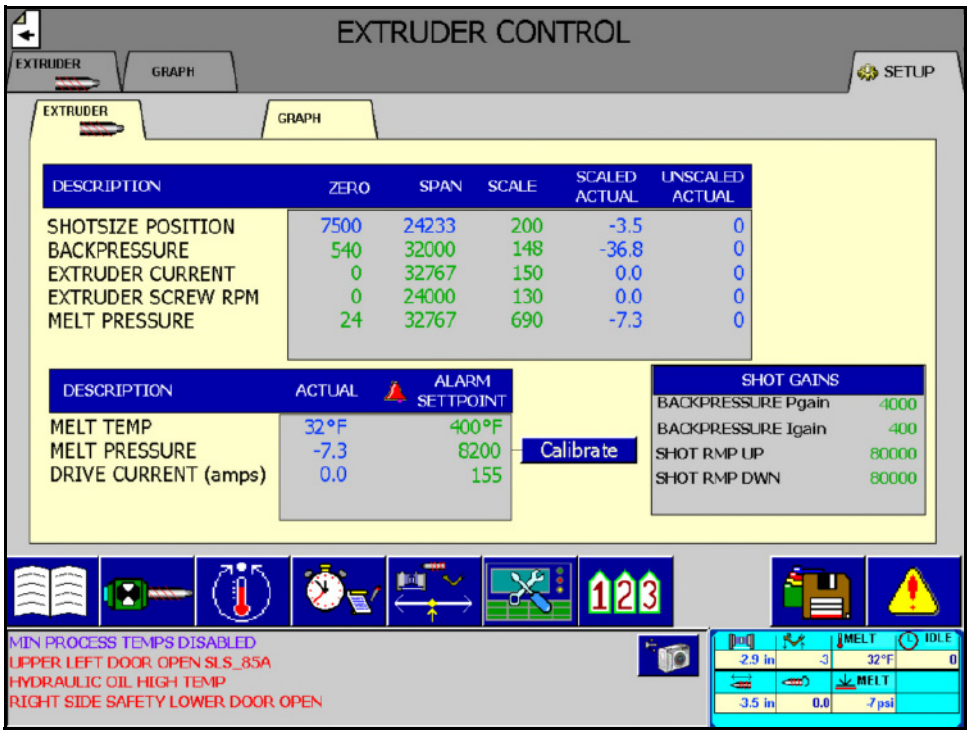
Calibration

Calibration establishes a range of motion for machine components. Calibration is performed after machine repairs or parts replacement. You can perform calibration for:

- Extruder
- Clamp and Swing Arms
- Parison Profile
- Temperature

Extruder Control - Calibration Screen

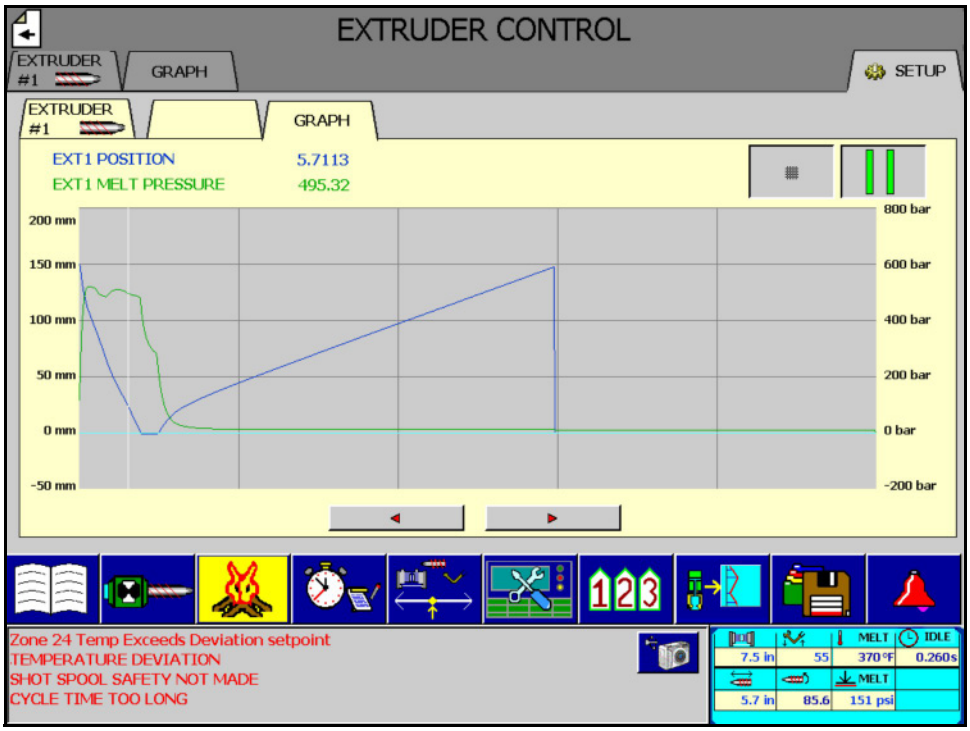
The Extruder Control - Calibration screen allows the user to calibrate machine parameters. Choose the area that needs to be calibrated. All calibration settings are factory set.



Extruder Control Screen

Extruder Control - Graph Screen

This screen displays the actual extruder position and extruder melt pressure graph. The button displays the grid, and pauses the trace.



Extruder Control - Graph

Performing Extruder Drive Calibration

To perform Drive Calibration:

a. Touch on the menu bar.

b. If prompted, enter Security Code.

c. From the Machine Setup and Calibration menu, touch DRIVE CALIBRATION.

d. Enter Melt Pressure:

- Check melt pressure with the machine idle for approximately 24 hours.
- Enter the amount as the Zero setpoint.
- Toggle PRESS CALIBRATE.
- Record the Unscaled Actual value for Melt Pressure.
- Enter the recorded value as the Span setpoint.
- Enter 80% of transducer range as Scale.

e. Toggle PRESS FOR MELT CALIBRATION to DISABLED.

f. Enter Extruder Current:

- Enter 0 for the Zero setpoint.
- Run the motor full out and record motor load (amps).
- Record the Unscaled Actual value for Extruder Current.
- Enter the recoded value as the Span setpoint.
- Enter the motor load (amps) as the Scale setpoint.

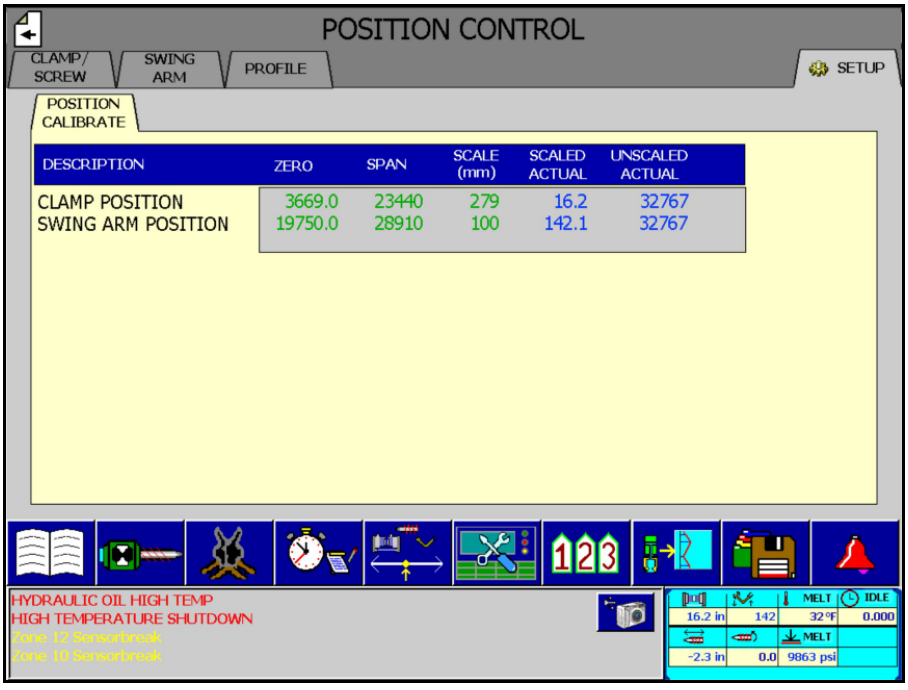
g. Enter Extruder Motor RPM:

- Enter 0 for the Zero setpoint.
- Run the extruder motor full out and record actual rpm.
- Enter the Unscaled Actual value as the Span setpoint.
- Enter the Actual motor rpm value as the Scale setpoint.

Note: *Extruder #2 is an optional feature. If the Recip machine has the Extruder #2 tab on the Extruder Control Screen, follow the same procedures as above in the topic "Performing Extruder Drive Calibration."*

Position Control - Position Calibrate Screen

The Position Control - Position Calibration screen allows the user to calibrate the clamp, and swing-arm(s). After calibration the user of the machine can enter setpoints for linear positions to run different products.



Position Control - Position Calibrate Screen

	Function
ZERO	Displays current zero position. Zero setpoint reflects the minimum signal from the drive at a stop condition.
SPAN	Displays current span position. Span setpoint reflects the maximum signal.
SCALE	Displays the engineering units that are viewed on processor.
SCALED ACTUAL	Displays actual engineering units.
UNSCALED ACTUAL	Displays the value of the signals in microprocessor units.

Performing Clamp Position Calibration

To perform clamp position calibration:

a. Touch  on the menu bar.

b. If prompted, enter Security Code.

c. From the Machine Setup and Calibration menu, touch POSITION CALIBRATION.

d. Place the clamp in the closed position.

e. Record the Unscaled Actual value for Clamp Position.

f. Enter the recorded value as the Zero setpoint.

g. Place the clamp in the full open position.

h. Record the Unscaled Actual value for Clamp Position.

i. Enter the recorded value as the Span setpoint.

j. Measure the physical travel of the linear potentiometer.

k. Enter the measurement as the Scale setpoint.

Performing Screw Position Calibration

To perform screw position calibration:

a. Touch  on the menu bar.

b. If prompted, enter Security Code.

c. From the Machine Setup and Calibration menu, touch POSITION CALIBRATION.

d. Place the clamp in the closed position.

e. Record the Unscaled Actual value for Screw Position.

f. Enter the recorded value as the Zero setpoint.

g. Manually retract the screw.

h. Record the Unscaled Actual for Screw Position.

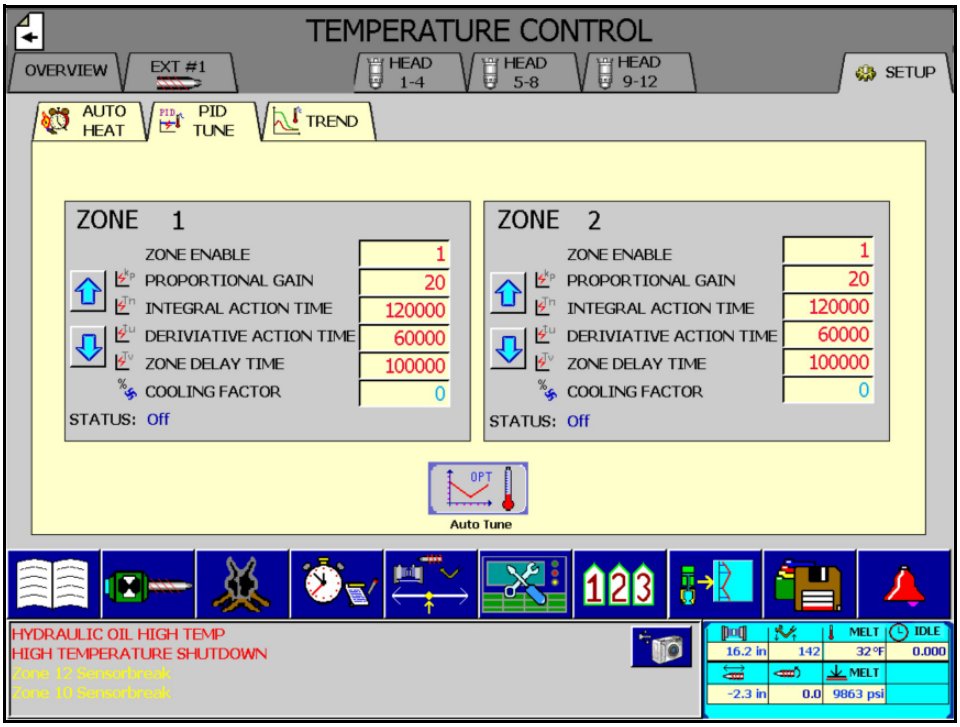
i. Enter the recorded value as the Span setpoint.

j. Measure the physical travel of the linear potentiometer.

k. Enter the measurement as the Scale setpoint.

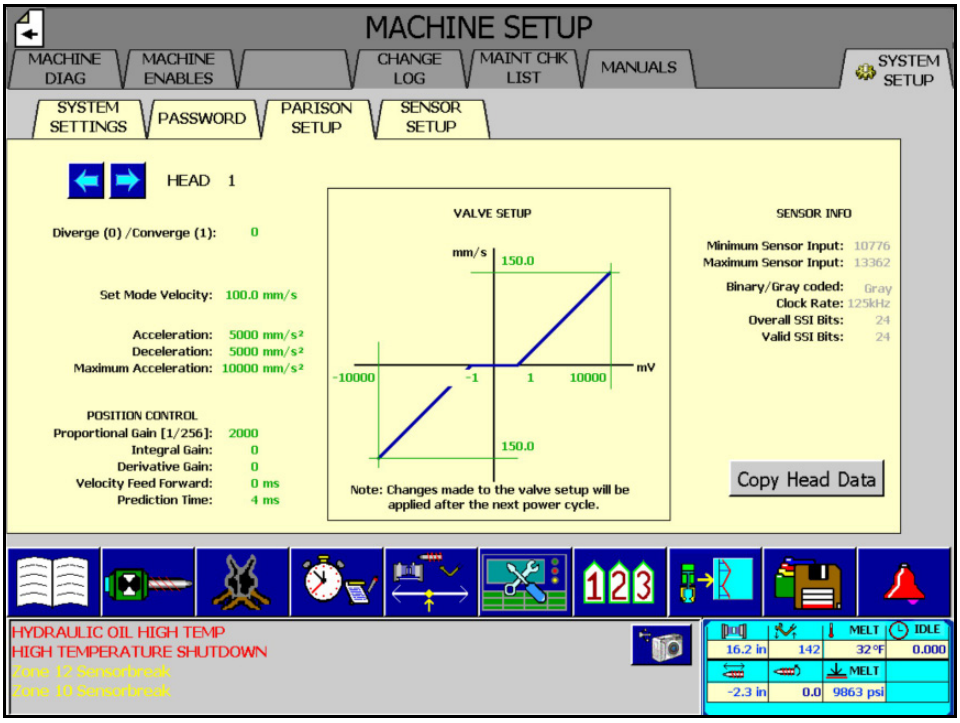
Manually Tuning the Machine Temperature Zone PID

An engineer typically adjusts the Machine Temperature Zone PID (Proportional Integral Derivative), to get the best temperature curve appropriate for the process. Temperature zones are defined on the Temperature Control screens.



Temperature Control - PID Tuning Screen

Machine Setup - Parison Calibration

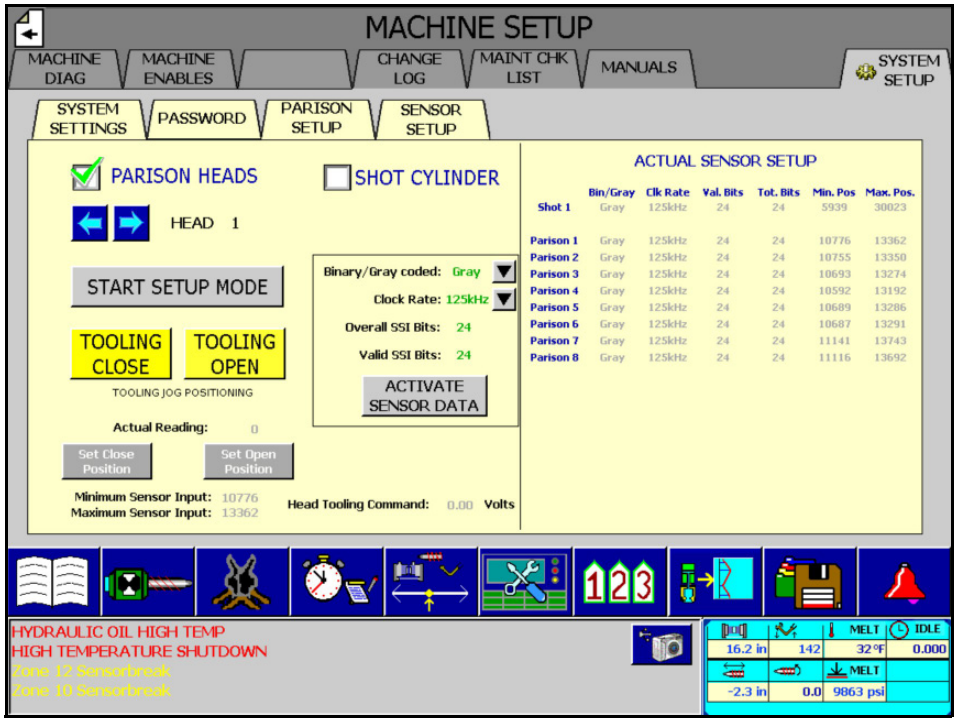


Machine Setup - Parison Calibration Screen

Using this screen, the fix-data for the parison controllers can be set. Notice that the changes of fix-data is not possible in semi, process or full automatic mode. Use the topmost input to choose the desired parison-controller.

	Function
DIVERGE / CONVERGE	Invert all parameters for invergent/divergent head.
SET MODE VELOCITY	Velocity in setting mode movement.
ACCELERATION	Acceleration rate for axis-movement (only for positioning).
DECELERATION	Deceleration rate for axis-movement (only for positioning).
MAX ACCELERATION	Maximum acceleration/deceleration during profile-movement.
VELOCITY FILTER TIME	Filter time for the velocity-measurement.
POSITION CONTROL	
PROPORTIONAL GAIN	Proportional gain of the position-controller (1/256).
INTEGRAL GAIN	Integral amplification.

	Function
VELOCITY FEED FORWARD	Kd - Gain for the velocity feed forward.
VELOCITY FEED FORWARD	Tf - Filter time for the velocity feed forward (derivative)
PREDICTION TIME	Prediction time for the velocity feed forward (time-based parison only)
VALVE SETUP	Voltages and velocities according to valve characteristics.



Machine Setup - Sensor Setup Screen

Calibrating Parison Heads

a. Select Parison Heads and select the head you want to calibrate.

b. Press Start Setup Mode.

c. Press Tooling Close, then wait for the movement to be completed.

d. Press Set Close Position to capture the sensor input.

e. Press Tooling Close again to turn Off the jog function.

f. Press Tooling Open, then wait for movement to be completed.

g. Press Set Open Position to capture the sensor input.

h. Press Tooling Open again to turn Off the jog function.

After calibrating each direction, the captured data will be displayed on the right hand side of the screen for each head.

i. Press Exit Setup Mode to end the calibration procedure.

Calibrating the Shot Cylinder

a. Select Shot cylinder.

b. Press Start Setup Mode

With the shot cylinder all the way forward (empty):

c. Press Set Close Position to capture the sensor input.

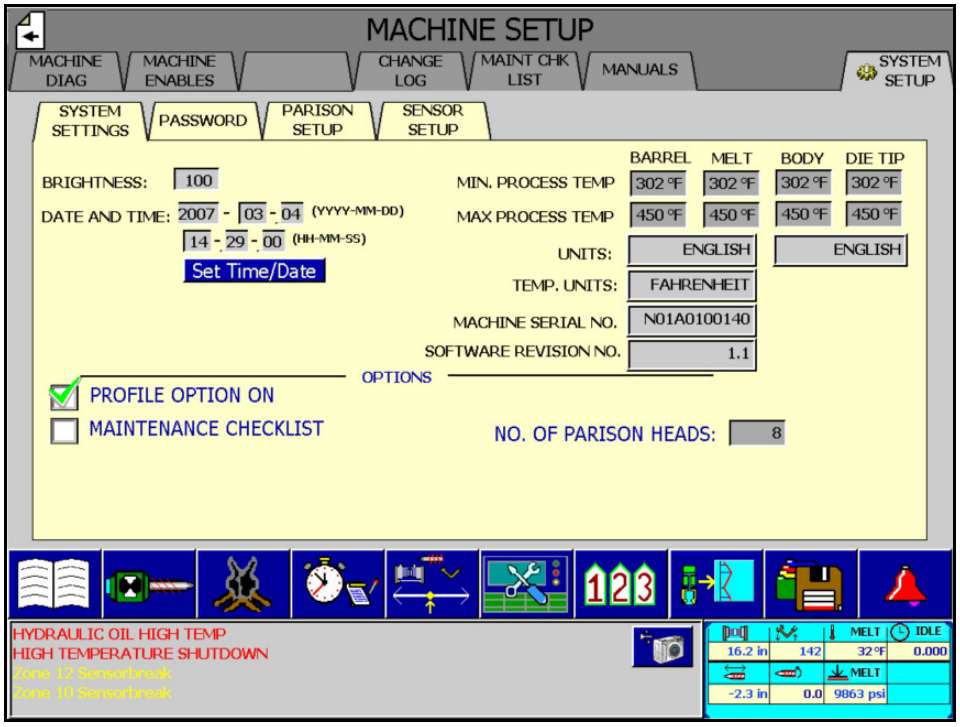
d. Press the Charge PB and wait for the shot cylinder to retract all the way back.

e. Press Set Open Position to capture the sensor input.

After calibrating each direction, the captured data will be displayed on the right hand side of the screen for the shot cylinder.

f. Press Exit Setup Mode to end the calibration procedure.

System Setup Screen



System Setup Screen

The parameters on the System Setup screen should only be set by authorized personnel.

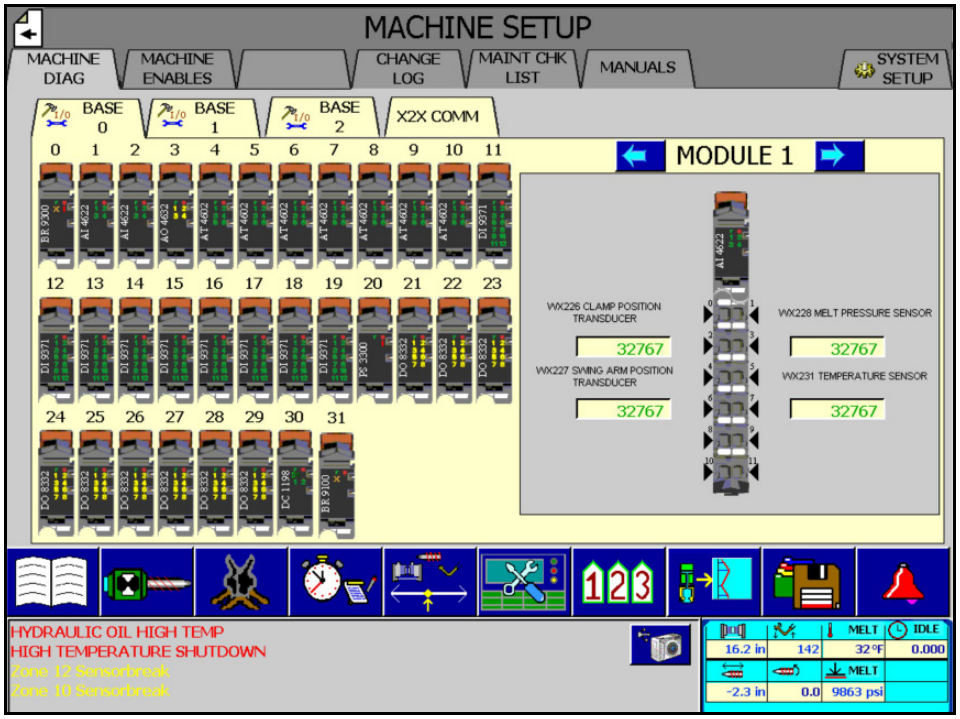
	Function
MIN/MAX PROCESS TEMP	Reset temperature settings. Set a temperature safety window for the barrel, heads and melt. If the actual temperatures are outside of these limits, an alarm will sound and the machine will stop.
UNITS	Change from English to metric.
ENGLISH	Change language (optional).
TEMP UNITS	Change temperature display from oF to oC.
MACHINE SERIAL NUMBER	Serial number of the machine.
SOFTWARE REV	Current software release number.
PROFILE OPTION	Option to Enable Parison programming (On or Off).
MAINT. CHECKLIST	This function is optional.
NO. OF PARISONS	Sets the number of active heads.

Monitoring Machine Diagnostics

The Machine Setup - Machine Diag screen displays:

- Inputs
- Outputs

Access this screen by touching  on the menu bar then pressing Machine Diag on the Machine Setup screen.



Machine Setup - Machine Diag Screen

Inputs and Outputs

The Input/Output is separated into Base 0, Base 1 and Optional Base 2. Each base may contain inputs/outputs or both. Each input/output card in the base has an ID number which indicates its function. they are:

AI 4622 = ANALOG INPUT MODULE

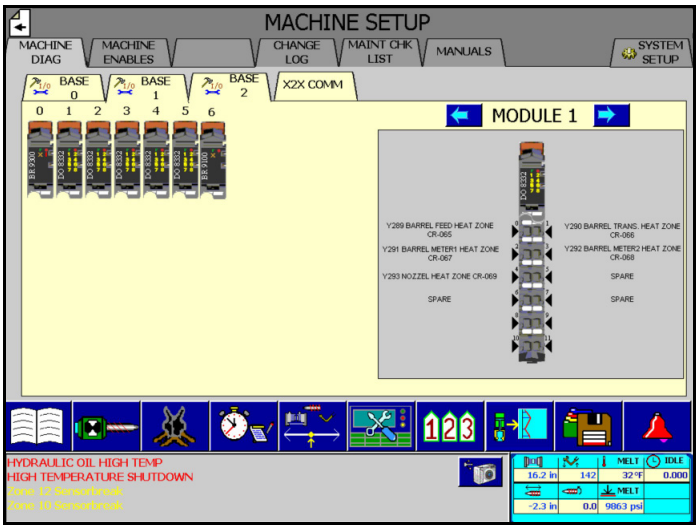
AO 4632 = ANALOG OUTPUT MODULE

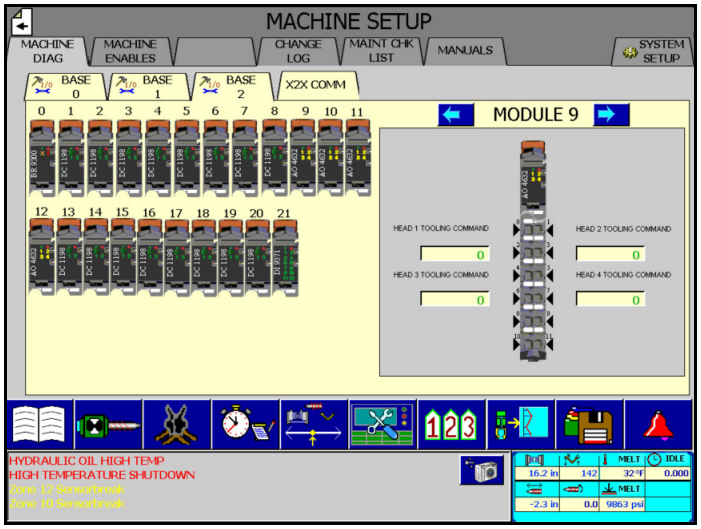
AT 6402 = THERMO COUPLE MODULE

D1 9371 = DIGITAL INPUT MODULE

DO 8332 = DIGITAL OUTPUT MODULE

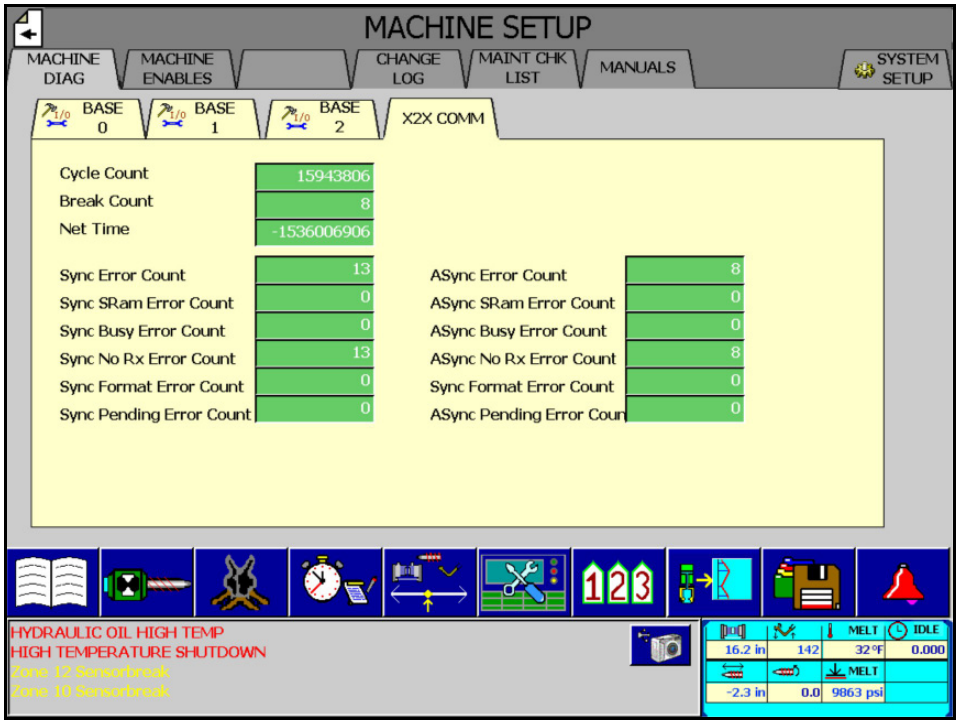
Pressing the module on the left side of the screen updates the module display on the right. Real time updates are displayed along with a description of the Input or Output.





Input and Output Diagnostics Screens (3-Layer shown)

X2X Communication Monitor



X2X Communication Monitor

The X2X Communication Monitor screen shows communication status of the communication from the operator station to the I/O. If the Sync or ASync error count increases, this may be an indication of a fault in the communications.